Amazon Prime TV Shows and Movies

Exploratory Data Analysis (EDA)

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving credits.csv to credits (1).csv
```

Project Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Amazon Prime Video dataset to understand trends and patterns in the available content. The analysis focuses on the distribution of movies and TV shows, release trends over the years, genre popularity, and IMDb ratings.

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving titles.csv to titles (1).csv
```

Dataset Description

This dataset contains information about titles available on Amazon Prime Video. It includes details such as title name, type of content (movie or show), release year, genres, runtime, IMDb ratings, and popularity scores. The dataset contains around 9,800 titles and multiple columns describing each title.

```
import os
os.listdir()
```

```
['.config',
 'titles (1).csv',
 'titles.csv',
 'credits (1).csv',
 'credits.csv',
 'sample_data']
```

Data Exploration

Basic data exploration was performed using Pandas functions such as head(), shape(), columns(), info(), and describe(). These functions helped understand the dataset structure, number of rows and columns, and the data types of each column.

```
import pandas as pd

titles = pd.read_csv('titles.csv')
credits = pd.read_csv('credits.csv')

titles.head()
```

	id	title	type	description	release_year	age_certification	runtime	genres	production_countries	seasons	im
0	ts20945	The Three Stooges	SHOW	The Three Stooges were an American vaudeville ...	1934	TV-PG	19	['comedy', 'family', 'animation', 'action', 'f...	['US']	26.0	tt08
1	tm19248	The General	MOVIE	During America's Civil War, Union spies steal ...	1926	NaN	78	['action', 'drama', 'war', 'western', 'comedy'...	['US']	NaN	tt00
2	tm82253	The Best Years of Our Lives	MOVIE	It's the hope that sustains the spirit of ever...	1946	NaN	171	['romance', 'war', 'drama']	['US']	NaN	tt00
3	tm83884	His Girl Friday	MOVIE	Hildy, the journalist former wife of newspaper...	1940	NaN	92	['comedy', 'drama', 'romance']	['US']	NaN	tt00
4	tm56584	In a Lonely Place	MOVIE	An aspiring actress begins to suspect that her...	1950	NaN	94	['thriller', 'drama', 'romance']	['US']	NaN	tt00

Handling Missing Values

The dataset contains some missing values in columns such as description, age certification, and seasons. Missing values were identified using the isnull().sum() function. These values were handled appropriately during the analysis process.

```
titles.shape
```

```
(9871, 15)
```

Key Insights

1. Movies are more common than TV shows on Amazon Prime.
2. The number of titles released has increased significantly in recent years.
3. Most titles have IMDb ratings between 6 and 7.
4. Genres such as drama, comedy, and action are among the most common.
5. Some titles have very high IMDb ratings, indicating strong audience appreciation.

```
titles.columns
```

```
Index(['id', 'title', 'type', 'description', 'release_year',
       'age_certification', 'runtime', 'genres', 'production_countries',
       'seasons', 'imdb_id', 'imdb_score', 'imdb_votes', 'tmdb_popularity',
       'tmdb_score'],
      dtype='object')
```

Conclusion

In this project, Exploratory Data Analysis was performed on the Amazon Prime Video dataset to identify patterns in content distribution, release trends, genres, and ratings. The analysis provides useful insights into the type of content available on the platform and how it has grown over time.

Double-click (or enter) to edit

```
titles.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9871 entries, 0 to 9870
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   id                    9871 non-null   object
 1   title                 9871 non-null   object
 2   type                  9871 non-null   object
 3   description           9752 non-null   object
 4   release_year          9871 non-null   int64
 5   age_certification     3384 non-null   object
 6   runtime               9871 non-null   int64
 7   genres                9871 non-null   object
 8   production_countries  9871 non-null   object
 9   seasons               1357 non-null   float64
 10  imdb_id               9204 non-null   object
 11  imdb_score            8850 non-null   float64
 12  imdb_votes            8840 non-null   float64
 13  tmdb_popularity       9324 non-null   float64
 14  tmdb_score            7789 non-null   float64
dtypes: float64(5), int64(2), object(8)
memory usage: 1.1+ MB
```

```
titles.isnull().sum()
```

	0
id	0
title	0
type	0
description	119
release_year	0
age_certification	6487
runtime	0
genres	0
production_countries	0
seasons	8514
imdb_id	667
imdb_score	1021
imdb_votes	1031
tmdb_popularity	547
tmdb_score	2082

dtype: int64

```
titles = titles.dropna(subset=['imdb_score'])
```

```
titles.isnull().sum()
```

	0
id	0
title	0
type	0
description	30
release_year	0
age_certification	5648
runtime	0
genres	0
production_countries	0
seasons	7684
imdb_id	0
imdb_score	0
imdb_votes	10
tmdb_popularity	328
tmdb_score	1530

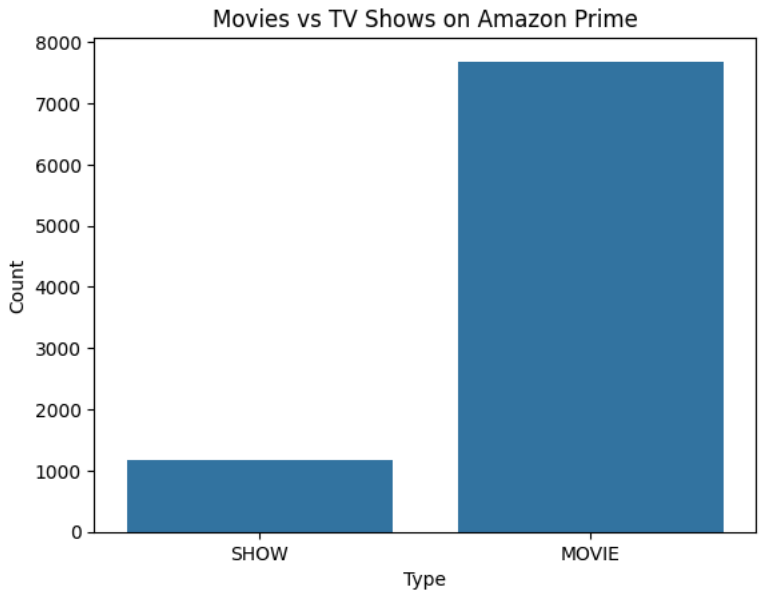
dtype: int64

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='type', data=titles)

plt.title("Movies vs TV Shows on Amazon Prime")
plt.xlabel("Type")
plt.ylabel("Count")

plt.show()
```



Visualization 1: Distribution of Movies vs TV Shows

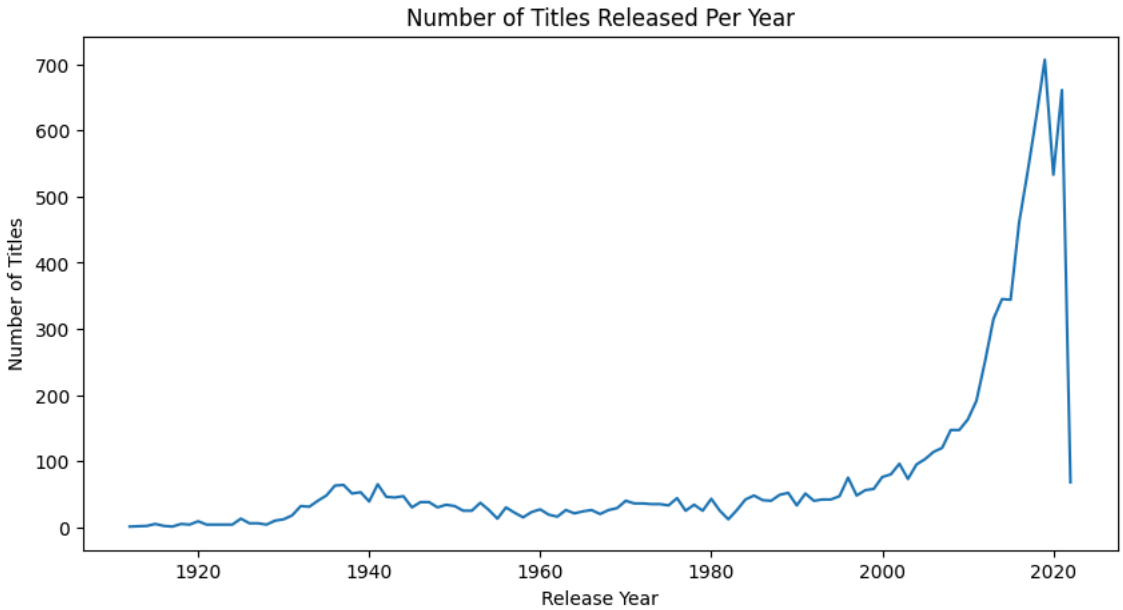
This bar chart shows the number of Movies and TV Shows available on Amazon Prime.

Insight: From the graph, we can see that Movies are significantly more numerous than TV Shows in the dataset. This indicates that Amazon Prime has a larger collection of movies compared to television series. The platform appears to focus more on providing movie content to its audience.

```
titles['release_year'].value_counts().sort_index().plot(figsize=(10,5))

plt.title("Number of Titles Released Per Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")

plt.show()
```



Visualization 2: Titles Released Per Year

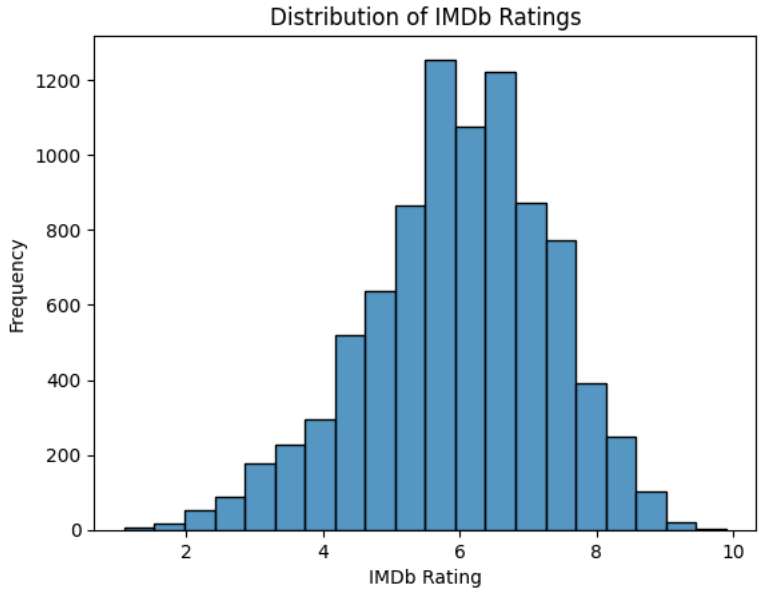
This line graph shows how many titles were released each year on Amazon Prime.

Insight: From the graph, we can observe that the number of titles released has increased significantly in recent years. This trend reflects the rapid growth of streaming platforms and the increasing demand for digital entertainment content worldwide.

```
sns.histplot(titles['imdb_score'], bins=20)

plt.title("Distribution of IMDb Ratings")
plt.xlabel("IMDb Rating")
plt.ylabel("Frequency")

plt.show()
```



Visualization 3: IMDb Rating Distribution

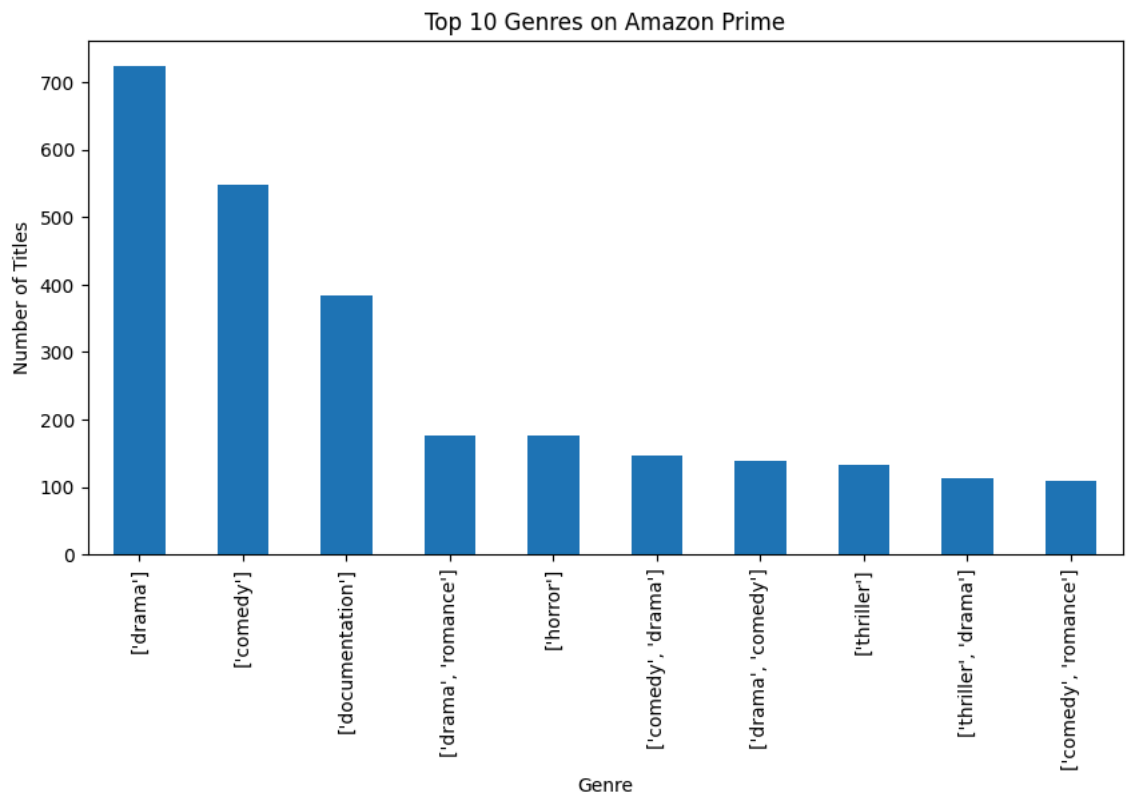
This histogram shows the distribution of IMDb ratings for titles available on Amazon Prime.

Insight: From the graph, we can observe that most titles have IMDb ratings between 6 and 7. This suggests that the majority of the content on Amazon Prime is moderately well-rated by viewers.

```
titles['genres'].value_counts().head(10).plot(kind='bar', figsize=(10,5))

plt.title("Top 10 Genres on Amazon Prime")
plt.xlabel("Genre")
plt.ylabel("Number of Titles")

plt.show()
```



Visualization 4: Top Genres on Amazon Prime

This bar chart shows the most common genres among titles available on Amazon Prime.

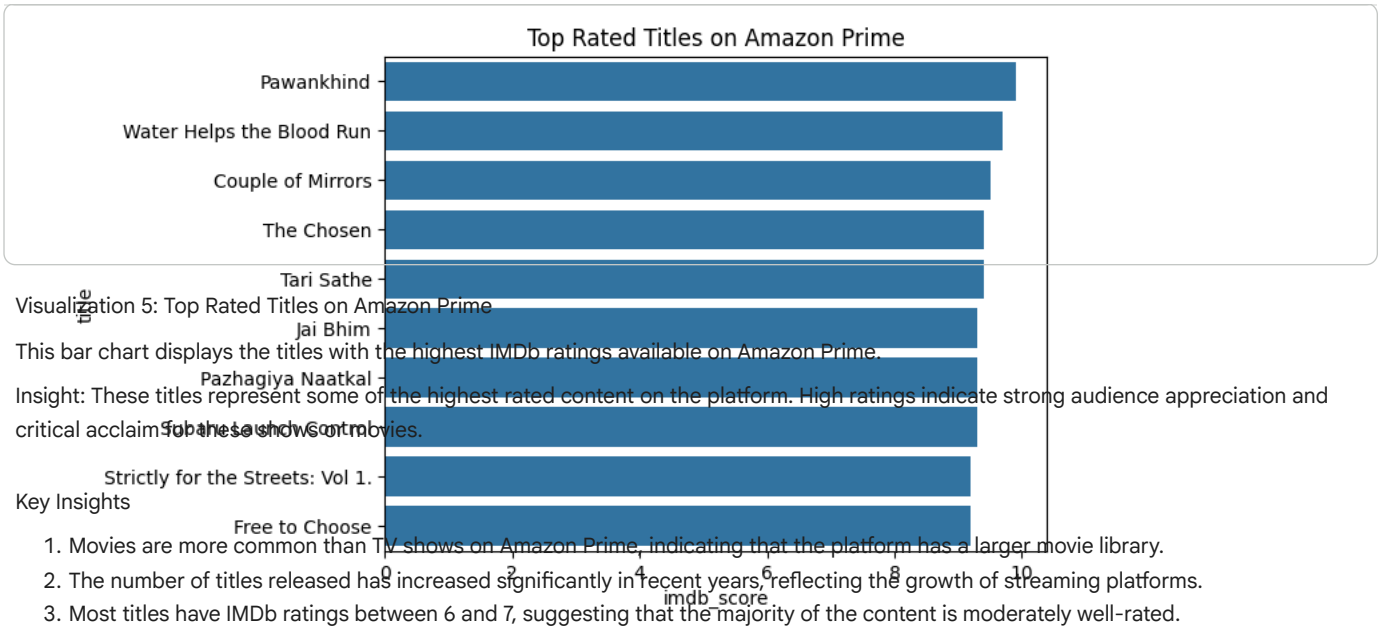
Insight: From the graph, we can see that certain genres appear more frequently than others. Genres such as drama, comedy, and action are among the most common. This indicates that these genres are popular among viewers and widely produced by content creators.

```
top_titles = titles.sort_values(by='imdb_score', ascending=False).head(10)

sns.barplot(x='imdb_score', y='title', data=top_titles)

plt.title("Top Rated Titles on Amazon Prime")

plt.show()
```



Visualization 5: Top Rated Titles on Amazon Prime

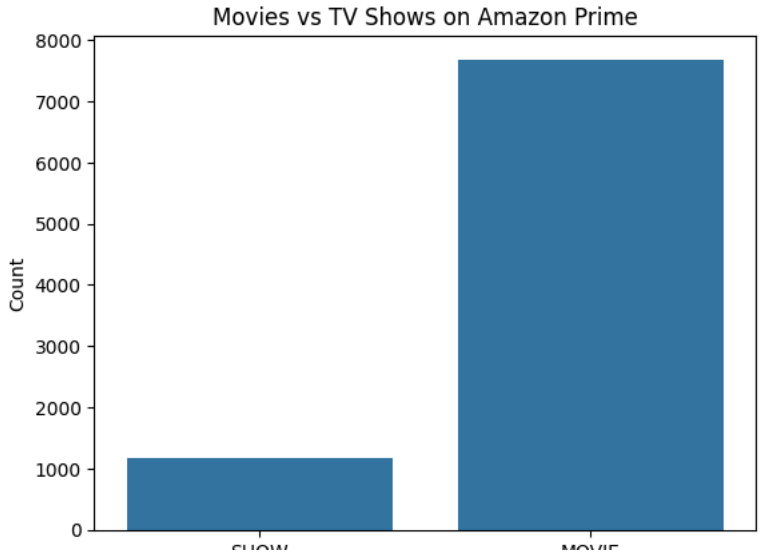
This bar chart displays the titles with the highest IMDb ratings available on Amazon Prime.

Insight: These titles represent some of the highest rated content on the platform. High ratings indicate strong audience appreciation and critical acclaim for these shows or movies.

Key Insights

1. Movies are more common than TV shows on Amazon Prime, indicating that the platform has a larger movie library.
2. The number of titles released has increased significantly in recent years, reflecting the growth of streaming platforms.
3. Most titles have IMDb ratings between 6 and 7, suggesting that the majority of the content is moderately well-rated.
4. Popular genres such as drama, comedy, and action appear frequently in the dataset.
5. Some titles have very high IMDb ratings, showing strong audience appreciation for certain content.

Conclusion

In this project, Exploratory Data Analysis (EDA) was performed on the Amazon Prime Video dataset. The analysis helped identify trends related to content type, release year, genre popularity, and ratings. The results show that movies dominate the platform, content production has increased over time, and certain genres are more common. These insights help understand the content strategy and audience preferences in the streaming industry.

```
# Visualization 1: Distribution of Movies vs TV Shows
sns.countplot(x='type', data=titles)

plt.title("Movies vs TV Shows on Amazon Prime")
plt.xlabel("Type")
plt.ylabel("Count")

plt.show()
```